

You should have two exam packets!

This is part 1 of 2

Name_____

Discussion Section Instructor_____

Discussion Section Number_____

Econ 302: Fall 2016 (Eudey)

First Exam, Version 1

Part 1 of 2

The exam has a total of 54 points. You have 70 minutes to complete the exam, so you must move very quickly.

Please carefully follow these instructions (*0.5 point is deducted from your total score if it is not completed correctly*):

- Write your name on your ScanTron
- Write the number of your discussion section under "Special Codes" spaces ABC of your ScanTron form and fill in the bubbles.
- Write the version number of your exam booklet under space D of your ScanTron form and fill in the bubbles.

Example: 3012 (discussion section number 301 + version number 2)

The exam is worth 54 points. There are 36 multiple choice questions each worth 1 point and short answer questions worth 18 points. The short answer in part 1 is worth 10 points.

Multiple Choice. Please enter your answer on the ScanTron form.

1. When a macroeconomic aggregate is procyclical
 - A) it grows faster than GDP.
 - B) its deviations from trend generally change before the deviations from trend in GDP do.
 - C) its deviations from trend generally change more than the deviations from trend in GDP.
 - D) its deviations from trend are more often of the same sign as the deviations from trend in GDP.

2. Average labor productivity tends to be
 - A) procyclical and less variable than real GDP.
 - B) procyclical and more variable than real GDP.
 - C) countercyclical and less variable than real GDP.
 - D) countercyclical and more variable than real GDP.

3. Real business cycle theory argues that the primary cause of business cycles is fluctuations in
 - A) preferences.
 - B) government spending.
 - C) the importance of externalities.
 - D) total factor productivity.

4. The weight of empirical evidence suggests that in the United States, the real wage rate is
 - A) acyclical.
 - B) bicyclical.
 - C) procyclical.
 - D) countercyclical.

5. In the one-period model, what do we assume about household preferences?
 - A) Households prefer more to less.
 - B) Households like money.
 - C) Households dislike taxes.
 - D) Households care about others.

6. A consumer is said to be indifferent between two consumption bundles

- A) when the consumer doesn't care about his or her consumption bundle.
- B) when the two bundles provide equal amounts of utility.
- C) when the consumer chooses the bundles equally often.
- D) when the consumer is indecisive.

7. According to our model it is likely that an increase in government spending

- A) increases consumption, increases hours worked, and increases the real wage.
- B) reduces consumption, increases hours worked, and increases the real wage.
- C) reduces consumption, increases hours worked, and reduces the real wage.
- D) reduces consumption, reduces hours worked, and reduces the real wage.

8. We assume that the representative consumer's preferences exhibit the properties that

- A) it is preferred to have more of one and less of another than more of everything
- B) more is always preferred to less and that each consumer has one strictly favorite good.
- C) each consumer has one strictly preferred good and that consumption and leisure are both normal goods.
- D) consumption and leisure are both normal goods and that the consumer likes diversity in his or her consumption bundle.

9. An increase in lump-sum taxes has the following impact on the budget constraint

- A) a parallel move down.
- B) a parallel move up.
- C) a tilting downward to the left.
- D) a tilting upward to the right.

10. A positive, pure income effect can be obtained by

- A) increasing the real wage.
- B) increasing the tax.
- C) increasing the dividend.
- D) decreasing leisure.

11. When the wage increases, the substitution effect in the household's choices leads to

- A) a decrease in consumption and leisure.
- B) a decrease in consumption and an increase in leisure.
- C) an increase in consumption and a decrease in leisure.
- D) an increase in consumption and leisure.

12. According to our model a shift in increase in preferences away from leisure is likely to result in

- A) increased consumption, increased hours worked, and increased real wage.
- B) increased consumption, increased hours worked, and increased real wage.
- C) reduced consumption, increased hours worked, and reduced real wage.
- D) reduced consumption, reduced hours worked, and reduced real wage.

13. Changes in government spending are not likely causes of business cycles because changes in government spending predict

- A) countercyclical real wages.
- B) procyclical real wages.
- C) countercyclical employment.
- D) procyclical employment.

14. In the production function, $Y = zF(K, Nd)$, total factor productivity is

- A) Y/K .
- B) Y/Nd .
- C) F/Y .
- D) z .

15. The Social Planner's solution is found where

- A) $MRS = w$
- B) $MRS = -U(I)/U(C)$
- C) $MRS = MRT$
- D) $MRT = z$.

16. The first fundamental theorem of welfare economics states that

- A) under certain conditions, a competitive equilibrium is Pareto optimal.
- B) a competitive equilibrium is always Pareto optimal.
- C) under certain conditions, a Pareto optimum is a competitive equilibrium.
- D) a Pareto optimum is always a competitive equilibrium.

17. When the representative firm maximizes profits

- A) production is at its maximum.
- B) the slope of the production function is at its flattest.
- C) labor costs are minimized.
- D) the marginal product of labor equals the wage.

18. In response to an increase in total factor productivity

- A) both the substitution effect and the income effect suggest that hours worked should increase.
- B) the substitution effect suggests that hours worked should increase, while the income effect suggests that hours worked should decrease.
- C) the substitution effect suggests that hours worked should decrease, while the income effect suggests that hours worked should increase.
- D) both the substitution effect and the income effect suggest that hours worked should decrease.

19. In a competitive equilibrium all these relationships must hold except one.
Which one?

- A) $N_d = N_s$
- B) $Y = G + C$
- C) $G = T$
- D) $w = z$

20. In the one-period competitive model we have been studying

- A) both consumption and total factor productivity are exogenous.
- B) consumption is exogenous and total factor productivity is endogenous.
- C) consumption is endogenous and total factor productivity is exogenous.
- D) both consumption and total factor productivity are endogenous.

21. A Pareto optimum requires all of the following except

- A) $MP_N = -\text{slope of } PPF$.
- B) $MRS_{1,C} = MRT_{1,C}$.
- C) $MRS_{1,C} = MP_N$.
- D) $MP_N = w$.

22. A competitive equilibrium

- A) is always economically efficient.
- B) is efficient only if there is an externality.
- C) is economically efficient only given some special conditions.
- D) does not exist without government taxation.

23. Under a Pareto Optimum

- A) it is always possible to improve someone's welfare.
- B) it is never possible to improve someone's welfare.
- C) one can only reduce someone's welfare.
- D) it is impossible to reduce someone's welfare.

24. A competitive equilibrium may fail to be Pareto optimal due to all of the following **except**

- A) inequality.
- B) externalities.
- C) distorting taxes.
- D) non-price-taking firms.

25. The presence of a distorting tax on wage income can result in

- A) $MP_N < MRT_{1,C}$.
- B) $MRT_{1,C} < MRS_{1,C}$.
- C) $MP_N < w$.
- D) $MRS_{1,C} < MP_N$

26. An increase in government spending shifts the PPF

- A) upward, but does not change its slope.
- B) upward, and also changes its slope.
- C) downward, but does not change its slope.
- D) downward, and also changes its slope.

27. Changes in total factor productivity are plausible causes of business cycles because

- A) of the welfare theorems.
- B) the U.S. government is following supply-side economic policy.
- C) the model matches many stylized facts.
- D) prices are countercyclical.

28.If the government replaces a lump sum tax with a proportional labor income tax, then

- A) employment and output increase.
- B) employment increases and output decreases.
- C) employment decreases and output increases.
- D) employment and output decrease.

29.The assumption of the model used in class is that when capital and technology increase

- A)They increase with labor demand
- B)They substitute for labor demand
- C)They increase because of an increase in labor demand
- D)They increase because of a decrease in labor demand

30.The competitive equilibrium is the most efficient allocation if

- A) there are no externalities
- B) there are no income taxes
- C) there is perfect information
- D) all of the above

31.With a linear production function in labor only, which of the following must be true?

- A) The representative household works as much as possible.
- B) The representative firm makes large profits.
- C) The real wage equals total factor productivity.
- D) The marginal product of labor exceeds the real wage.

32. At the competitive equilibrium with a positive proportional labor income tax
- A) the real wage after tax exceeds the marginal product of labor.
 - B) the real wage after tax equals the marginal product of labor.
 - C) the real wage after tax is lower than the marginal product of labor.
 - D) We cannot say.

33. Assume that the representative household has utility represented by $U = \alpha \ln C + \beta \ln I$. This implies that when real wages increase, we expect
- A) Income effects to dominate substitution effects and for employment to rise
 - B) Income effects to be dominated by substitution effects and for employment to rise
 - C) Income effects to exactly offset substitution effects and for employment to be unchanged
 - D) Income effects to dominate substitution effects and for employment to fall

34. Which of the following utility functions is most consistent with an observation that rich people in the country of Eudeyland work about the same amount as do the poor?
- A) $U = \alpha \ln C + \beta \ln I$
 - B) $U = \alpha C + \beta I$
 - C) $U = \alpha C + \beta \ln I$
 - D) $U = \alpha \ln C + \beta I$

35. In the production function $Y = zK^\alpha N^\beta$ constant returns to scale will be satisfied when
- A) $\alpha + \beta = 1$
 - B) α and $\beta = 1$
 - C) $\alpha + (1 - \alpha) = \beta$
 - D) $1 + \alpha = \beta$

36. In the model from class, an increase in the minimum wage would most likely result in

- A) Unemployment
- B) Less employment
- C) Lower MPN
- D) all of the above

Short Answer worth 10 points. Please enter your answer in the space below.

Assume the following model economy:

$$1) U = \alpha \ln C + \beta \ln I$$

$$2) Y = zN$$

$$3) C = wN(1-t) + \pi$$

$$4) h = N + I$$

$$5) \pi = Y - wN$$

where the variables have the same meanings and definitions that we've used throughout the term.

- I. (8 points) What is the market (competitive) equilibrium value for leisure in this model? **Show your work** (*Hint: most of the points come from how you solve the problem not whether your final solution is correct.*)

- II. (2 points) Would the Social Planner's solution to the same problem you did in question 1 result in a higher or lower level of employment? Explain why that is, without doing any math.

Go on to Part 2 of the Exam